多莉：一只羊的复制

Dolly, the Copied Sheep



白脸小羊为何由黑脸母羊生下?

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英文原文

Dolly was born at Roslin, Scotland, on 5 July 1996. Her genome came from a Finn Dorset ewe's mammary cell, fused with a Blackface egg stripped of its nucleus. A Blackface surrogate carried the embryo through pregnancy, yet her white face was proof of a genetically different lineage. She alone survived from 277 attempts, an extraordinary failure rate.

Dolly is popularly called the first clone, a misconception. Her predecessors included frogs cloned from transplanted nuclei in 1962 and Roslin's embryo-derived duo, Megan and Morag, in 1995. She was genuinely the first mammal cloned from an adult cell. Roslin's announcement on 22 February 1997 coincided with the paper, and her name, a mammary pun, honoured Dolly Parton.

The news provoked uproar and apprehension about the ethics of human cloning. In March 1997 Clinton prohibited federal funding for such experiments. In 1999 her telomeres, or chromosome tips, were reported shortened by about 20% for her age. That fed anxiety; some speculated she was ageing prematurely.

Dolly was put down, aged six, on 14 February 2003, with arthritis and a progressive lung disease caused by a sheep virus. Indoor confinement probably contributed. In July 2016, 13 clones, four from her cell line, were ageing normally, without hypertension or diabetes. In November 2017, her skeletal X-rays showed ordinary arthritis, and fears of early onset were judged unfounded.

She bore six lambs by natural mating; Bonnie, first of these descendants, arrived in April 1998. Since 2003 her stuffed body, now famously an icon and memorial, has been encased in the National Museum of Scotland. Behind glass, her white face plainly testifies to her Finn Dorset ancestry.

中文译文

多莉于1996年7月5日在苏格兰罗斯林出生。她的基因组来自一只芬兰多塞特母羊的乳腺细胞；研究人员把这个细胞与一枚已去除细胞核的黑脸羊卵细胞融合。胚胎由一只黑脸羊代孕并孕育至出生，但多莉的白脸显示，她的遗传血统与代孕母羊不同。277次尝试中，只有她存活下来，失败率高得惊人。

人们常把多莉称为第一个克隆体，这是误解。在她之前，已有1962年通过移植细胞核克隆的青蛙，以及罗斯林1995年用胚胎细胞克隆出的梅根和莫拉格两只羊。她真正创下的纪录，是成为首只由成年细胞克隆的哺乳动物。罗斯林于1997年2月22日公布消息，相关论文也同时发表。她的名字带有乳腺相关的双关意味，向多莉·帕顿致意。

消息引起轩然大波，也引发人们对克隆人伦理的忧虑。1997年3月，克林顿禁止以联邦资金资助这类实验。1999年有报告称，多莉的端粒，即染色体末端，比她这个年龄应有的长度短约20%。这加重了焦虑，有人推测她正在过早衰老。

2003年2月14日，六岁的多莉被实施安乐死。她当时患有关节炎，以及由一种绵羊病毒引起、不断恶化的肺病；长期圈养在室内可能也是促成病情的因素。2016年7月，一项研究中的13只克隆羊都在正常衰老，没有高血压或糖尿病，其中4只来自与多莉相同的细胞系。2017年11月，对多莉骨骼X光片的检查显示，她患的是普通关节炎，对过早发病的担忧被认为缺乏根据。

多莉通过自然交配生下六只小羊。第一个后代邦妮于1998年4月出生。自2003年起，她的填充标本一直陈列在苏格兰国家博物馆的玻璃展柜中，如今已是广为人知的象征和纪念物。隔着玻璃，她的白脸依然清楚地表明其芬兰多塞特祖源。

“基因组来自乳腺细胞”主要指细胞核中的遗传物质；去核卵细胞仍保留自身的线粒体遗传物质，代孕母羊则负责孕育胚胎。

核心词汇

先记这22项。连着下方的原文短语一起回想。

genome n.
基因组
Her genome came

fuse with 动词短语
与……融合
fused with a Blackface egg

strip of 动词短语
去除；剥去
stripped of its nucleus

nucleus n.
细胞核
stripped of its nucleus

surrogate n.
代孕母羊；代孕者
Blackface surrogate

embryo n.
胚胎
carried the embryo through pregnancy

lineage n.
血统；世系
a genetically different lineage

clone n.
克隆体
the first clone

misconception n.
误解；错误观念
a misconception

mammal n.
哺乳动物
the first mammal cloned

coincide with 动词短语
与……同时发生
coincided with the paper

provoke v.
引起；激起
The news provoked uproar

apprehension n.
忧虑；担心
apprehension about the ethics

ethics n.
伦理；道德规范
the ethics of human cloning

prohibit v.
禁止
Clinton prohibited federal funding

speculate v.
推测
some speculated

prematurely adv.
过早地
she was ageing prematurely

progressive adj.
病情逐渐加重的
a progressive lung disease

confinement n.
限制活动；圈养
Indoor confinement

onset n.
发病；开始出现
fears of early onset

unfounded adj.
没有根据的
were judged unfounded

ancestry n.
祖源；世系
her Finn Dorset ancestry

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

Scotland 专有名词
苏格兰

pregnancy n.
孕期；怀孕

proof of 名词短语
……的证据

genetically adv.
从遗传上说

extraordinary adj.
异乎寻常的

popularly adv.
通常；普遍地

predecessor n.
先前的同类；前身

transplant v.
移植

duo n.
一对；两个一组

genuinely adv.
真正地；确实

announcement n.
公告；宣布

pun n.
双关语

honour v.
向……致意

uproar n.
轩然大波

chromosome n.
染色体

shorten v.
缩短

anxiety n.
焦虑

put down 动词短语
对动物实施安乐死

arthritis n.
关节炎

lung n.
肺

virus n.
病毒

contribute v.
是促成因素之一

hypertension n.
高血压

diabetes n.
糖尿病

skeletal adj.
骨骼的

mating n.
交配

descendant n.
后代

stuffed adj.
制成填充标本

famously adv.
众所周知地

icon n.
象征；标志性事物

memorial n.
纪念物

encase in 动词短语
封装在……里面

plainly adv.
清楚地

testify to 动词短语
表明；证明

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Dolly is popularly called the first clone, a _____.

02 Her _____ came from a Finn Dorset ewe's mammary cell

03 some speculated she was ageing _____.

04 fears of early onset were judged _____.

文章主线

- 诞生 研究者将去掉细胞核的卵细胞与成年母羊乳腺细胞融合，培育胚胎，1996年由代孕母羊产下多莉。
- 突破 277次尝试仅多莉存活；她是首只由成年细胞克隆的哺乳动物，此前已能用胚胎细胞克隆羊。
- 争议 成果引发人类克隆伦理争论；多莉的端粒（染色体末端）比同龄羊短约20%，又引起早衰猜测。
- 离世 多莉自然交配育下六只小羊；患有关节炎和病毒性肺病，六岁时被实施安乐死，不能据此认定克隆导致早衰。
- 复核 后续研究中13只克隆羊正常衰老；多莉骨骼复查也未支持关节炎过早出现的担忧。

参考答案　1 misconception 2 genome 3 prematurely 4 unfounded